

Regulator EMI Filter Design Choices and Values

Circuit Theory:

Voltage regulators are devices that can be used to create a steady output of power from an unregulated source, thereby protecting connected devices. They work by implementing a transistor in series with the voltage load to drop the excess and maintain a steady output. But this introduces more entry points for noise within the device, which is why, oftentimes, multiple EMI filters are necessary. Mainly, an input and output filter.

The battery rail connecting to the voltage regulator will introduce noise, so we will place an EMI filter there. A lot of components will be connected to the BMS, so this eliminates unwanted noise from every component. This mainly prevents erratic switching and any disturbances that may occur. The output filter solves the opposite issue from the input. Now, the voltage regulator is the source of the noise, meaning we must protect the components connected to this part.

To remove an adequate amount of noise, I have decided to use a standard pi-filter, also known as a C-L-C filter, because of its multiple stages of attenuation. The first capacitor eliminates any high-frequency noise; the inductor will block differential noise that makes it past the first capacitor; and the last capacitor should eliminate any remaining noise. Because the pi-filter is a third-order filter, the signal roll-off will be sharper, resulting in more dramatic noise removal.

To ensure the stability of the input voltage and to obtain a small DC resistance, the inductor must be as small as possible. This allows us to freely select a capacitor value, and then account for the inductor last.

As for the output filter, I have decided to place a 600Ω ferrite bead instead of an inductor. Ferrite beads are special in that they only absorb high-frequency signals and dissipate them as heat. There will be less noise coming out of the voltage regulator than in, meaning we don't necessarily need a C-L-C configuration.

Lastly, we must account for any resonance within the circuit. While this isn't as large an issue compared to the others, adding a resistor in series after the inductor will ensure this won't become an issue in the future.

Value Selection:

First, we will look at the construction of the first filter. Looking at the voltage regulator, we can see that the switching frequency has been set based on the RT pin (pin 3). There is a resistor connected to it with a value of $18.2k\Omega$.

$$f_{sw} = \frac{50 * 10^9}{18200} = 2.75MHz$$

$$f_c = \frac{2.75MHz}{10} = 275kHz$$

Based on the resistor, we should set our switching frequency at 275kHz in order to be one decade below the cutoff frequency. Then, based on this switching and cutoff frequency, we can find the minimum capacitance we need to achieve a reactance of 1.

$$X_c = \frac{1}{2\pi C f_c} \Rightarrow C = \frac{1}{2\pi X_c f_c}$$

$$C = \frac{1}{2\pi * 275kHz * 1\Omega} \Rightarrow C = \frac{1}{1727875.959} \Rightarrow C = 0.578\mu F$$

This means that at each node we will need a capacitance of around $0.578\mu\text{F}$ to have a reactance of 1Ω at 275kHz . But by adding a small $10\mu\text{F}$ to each capacitor, we can eliminate the low-end frequency as well. Doing this will give us a new reactance value.

$$X_c = \frac{1}{2\pi * 275\text{kHz} * 10.5\mu\text{F}} \Rightarrow X_c = \frac{1}{18.14269757} \Rightarrow X_c = 0.005\Omega$$

This is much lower than the initial estimate for reactance, which means that the circuit is doing a good job at cutting out noise. As for the second capacitor, I will place an identical capacitor setup.

All that's left for the input filter is to find the value for our inductor.

$$f_c = \frac{1}{2\pi\sqrt{LC}} \Rightarrow L = \frac{1}{(2\pi f_c)^2 C} \Rightarrow L = \frac{1}{(2\pi * 275\text{kHz})^2 * 4.7\mu\text{F}}$$

$$L = \frac{1}{14032110.06} \Rightarrow L = 71.26\text{nH}$$

This value is the theoretical minimum value we can place for our specific circuit. While this may work, it is preferable to round up the value to give us some room for error when the signal attenuates. For this reason, I have selected our inductance value to be $1\mu\text{H}$, as not only is it larger than what we need, giving us that space, but it is a common value for a real-world inductor.

For the output filter, all of the calculations are identical, but we don't need to be as strict for noise reduction. Much of the noise will have already been blocked by the input signal, and the voltage regulator itself won't introduce as much.

Given the fact that I have chosen different values than those calculated for our components, the actual cutoff frequency is changed.

$$f_c = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1\mu H * 4.7\mu F}} = \frac{1}{1.36216211 * 10^{-5}} \cong 73.4kHz$$

Just to confirm how much our filter cuts off, we can calculate attenuation to make sure it fits our needs.

$$Signal\ Attenuation = 40 \log_{10}\left(\frac{f_{sw}}{f_c}\right) = 40 \log_{10}\left(\frac{2.75MHz}{73.4kHz}\right) = 40 * 1.573 \cong 63dB$$

This means that the noise at 2.75MHz is greatly reduced by a factor of almost 200.